THREAT ADVISORY REPORT

IRGC-Linked Threat Actor Cluster - Middle East Operations

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01 Executive Summary

5	47	11	6	HIGH
APT CLUSTERS	KNOWN IOCs	MITRE TTPs	CVEs TRACKED	THREAT LEVEL

The Islamic Revolutionary Guard Corps (IRGC) and its affiliated cyber units continue to represent the most persistent and sophisticated state-sponsored threat to Middle East governments, critical infrastructure, defence contractors, and telecommunications providers as of April 2026. This advisory consolidates intelligence from 15+ primary sources - including CISA, FBI, Mandiant, ESET, Trend Micro, Picus Security, Tenable, and SecurityScorecard - covering five distinct IRGC-aligned threat clusters: APT33 (Elfin/Refined Kitten), APT34/OilRig (Earth Simnavaz), APT35/Charming Kitten (Mint Sandstorm), MuddyWater (Mango Sandstorm/TA450), and CyberAv3ngers (Shahid Kaveh Group/Bauxite). A sixth emerging cluster, UNC1549/Crimson Sandstorm, focused on aerospace and defence targets in Israel, UAE, and Turkey, is assessed with MEDIUM confidence.

The threat landscape has escalated significantly in 2025–2026, driven by the June 2025 twelve-day Iran–Israel kinetic conflict, subsequent U.S.-Israel strikes on IRGC facilities in February 2026, and Iranian retaliatory missile operations targeting GCC states. IRGC cyber operations are no longer solely espionage-focused - they now constitute a hybrid offensive combining intelligence collection, destructive ICS/OT attacks, influence operations, and hacktivist proxies as force multipliers. Organizations in the GCC, Jordan, Egypt, and Turkey face elevated risk from both direct APT intrusions and second-order disruption through supply chain compromise.

What Happened:
IRGC-aligned APTs have maintained persistent footholds in Middle Eastern critical infrastructure since early 2025 via credential theft and VPN compromise (CloudSEK, March 2026). CyberAv3ngers escalated from PLC defacement to deploying the IOCONTROL ICS malware platform and actively exploiting CVE-2021-22681 (Rockwell Automation authentication bypass) - confirmed by the six-agency CISA advisory AA26-097A on April 7, 2026.

Who Is Affected:
Government ministries, energy and water utilities, defence contractors, aviation, telecommunications, and healthcare organizations across Israel, Saudi Arabia, UAE, Qatar, Jordan, Egypt, Turkey, and Iraq. Second-order risk extends to European and South Asian firms with operational presence or supplier relationships in the region.

How Urgent:

CRITICAL. Multiple CVEs actively exploited in the wild with no patches available for CVE-2021-22681. CyberAv3ngers' ICS techniques have proliferated to an estimated 60+ affiliated groups (Tenable Apr 2026; single commercial source). Iranian retaliatory cyber posture following February 2026 strikes is assessed as ELEVATED. Immediate defensive action required.

What To Do:

- Immediately audit all internet-exposed PLCs, HMIs, and OT devices for default credentials (CVE-2021-22681, Unitronics Vison Series)
- Enforce MFA on all remote access, VPN endpoints, and Microsoft 365 / Entra ID accounts - APT33 password spraying is ongoing
- Deploy the Sigma rules in Section 11 to SIEM platforms; validate against Sysmon EID 1, 3, and 11 event coverage
- Review Section 10 IOCs against SIEM, EDR, and DNS/proxy logs immediately - 47 IOCs with provenance tags provided
- Activate IR pre-positioning protocols - notify legal/DPO teams of potential GDPR/DPDP notification obligations
- Hunt for BugSleep/MuddyRot, STEALHOOK, MINIBIKE, IOCONTROL indicators using pivots in Section 12

Analytical Confidence: *HIGH (multi-source corroboration across government advisories, commercial threat intelligence, and independent researcher analysis). Attribution to IRGC affiliation relies on CISA/FBI/DOJ indictments, U.S. Treasury sanctions designations, and Mandiant/ESET/Trend Micro technical research. Single-source dependency disclosure: UNC1549 section relies primarily on Mandiant/Google Cloud intelligence.*

02
Scope Statement & Applicability Filter

Scope: This advisory covers IRGC-affiliated and IRGC-directed threat actor clusters conducting operations against Middle East-region targets from January 2024 through April 15, 2026 (temporal cutoff). Coverage includes: APT33, APT34/OilRig, APT35/Charming Kitten, MuddyWater, CyberAv3ngers, and UNC1549. Nation-state Israeli offensive operations (Predatory Sparrow, Unit 8200) are outside scope. Opportunistic ransomware groups without confirmed IRGC nexus are outside scope.

What Is Not Covered: Historical operations prior to 2024; MOIS-directed campaigns without IRGC overlap; domestic Iranian cyber repression activity; Islamic Jihad or Hezbollah cyber infrastructure where IRGC involvement cannot be confirmed.

THREAT APPLICABILITY DECISION FRAMEWORK

Assess your organization against each question. Priority level determines the urgency of action required.

Question	If YES	If NO	Priority
Do you operate in or have significant commercial exposure to the Middle East?	Continue assessment - all remaining questions apply	Low direct risk; monitor for supply-chain and second-order threats only	BASELINE
Do you operate internet-exposed OT/ICS devices (PLCs, HMIs, SCADA)?	CRITICAL - apply Section 13 OT mitigations immediately. CVE-2021-22681 has no patch.	Lower ICS exposure; still apply IT-layer controls	CRITICAL
Do you use Microsoft 365, Azure, or Entra ID for identity management?	HIGH - enable Entra ID sign-in risk policies immediately; APT33 TOR spray is ongoing	Assess equivalent identity platform for credential-spray exposure	HIGH
Are you in defence, aerospace, aviation, or government sectors?	HIGH - UNC1549 and APT33 actively targeting these sectors; review Section 12 Pivot 06	Reduced targeting priority for these specific clusters	HIGH
Do you rely on Israeli-made technology or Israeli IT providers in your supply chain?	MEDIUM-HIGH - CyberAv3ngers and MuddyWater (Rashim breach) have demonstrated supply-chain targeting	Standard third-party risk controls apply	MEDIUM

Are you a financial institution or energy company in GCC states?	HIGH - OilRig/APT34 has a decade-long sustained focus on Gulf energy and financial institutions	Lower sector-specific targeting risk from APT34	**HIGH**

03
Threat Landscape Overview

Iran's offensive cyber ecosystem is bifurcated between the IRGC (military, directly under the Supreme Leader) and MOIS (Ministry of Intelligence, under civilian presidency). Both operate APTs with shared TTPs, operational coordination, and proxy hacktivist networks. The IRGC directly controls APT33, APT35, and CyberAv3ngers; MOIS controls APT34/OilRig and MuddyWater (CSIS, 2025). Since 2022, the boundary has blurred: leaked operational records (KittenBusters, GitHub, September–October 2025) revealed that IRGC Unit 1500 manages MuddyWater-adjacent operations through a network of front companies from the Shuhada base in Tehran (The Hacker News, December 2025).

SOURCE DISTRIBUTION

Intelligence Source	Coverage	Recency	Weight
CISA/FBI Joint Advisories (AA26-097A, AA23-335A)	CyberAv3ngers, ICS/OT	April 2026	CRITICAL
DOJ Indictment (Sep 2024)	APT35/IRGC hack-and-leak	Sep 2024	HIGH
Mandiant/Google Cloud (UNC1549)	Aerospace/Defence ME	Nov 2025	HIGH
ESET Research (MuddyViper)	MuddyWater, Israel/Egypt	Dec 2025	HIGH
Trend Micro (Earth Simnavaz)	APT34 Middle East	Oct 2024	HIGH
Picus Security	All Iranian APTs, TTPs	April 2026	HIGH
Nozomi Networks Telemetry	OT/ICS, Mar 2026	Mar 2026	HIGH
Tenable Research (CyberAv3ngers)	ICS/PLC escalation	April 2026	HIGH
CloudSEK ME Escalation Report	Regional threat context	Mar 2026	MEDIUM
SecurityScorecard (12-Day War)	Hacktivist IRGC nexus	Aug 2025	MEDIUM
CSIS Strategic Technologies Blog	Iran cyber ecosystem	2025	MEDIUM
Resecurity (IRGC targeting)	IRGC phishing campaigns	Sep 2024	MEDIUM
ReliaQuest (Iran-Israel)	APT34, APT35, CyberAv3ngers	Jun 2025	MEDIUM
HSToday (Pyroxene/Bauxite)	ICS kill-chain ops	Mar 2026	MEDIUM
CPJ (IRGC journalist targeting)	Banished Kitten/Handala	Jul 2025	MEDIUM

HISTORICAL COMPARISON

Methodology: Year-over-year comparison uses Nozomi Networks telemetry spike data (H2 2025 vs H1 2025) and Tenable/CISA advisory counts. Percentages reflect proportional change in confirmed detections, not absolute incident counts.

IRGC-affiliated activity escalated measurably across 2024–2026. Manufacturing and Transportation sectors were most targeted in March 2026 following the February U.S.-Israel strikes (Nozomi Networks, March 2026). CyberAv3ngers' ICS exploitation techniques have proliferated to an estimated 60+ affiliated groups since 2023 (Tenable Apr 2026; single commercial source -- CISA AA26-097A confirms broad proliferation without quantifying) (Tenable, April 2026), representing a qualitative capability diffusion that quantitative comparisons cannot fully capture. The June 2025 twelve-day conflict demonstrated for the first time Iran's ability to coordinate hundreds of hacktivist groups simultaneously with state APT operations against a single adversary (SecurityScorecard, August 2025; CSIS, 2025).

04
Attribution Assessment

Confidence Level: HIGH - Attribution to IRGC and MOIS nexus is supported by: (1) U.S. Treasury sanctions designations (February 2024) naming six IRGC-CEC officials directing CyberAv3ngers; (2) DOJ criminal indictment (September 2024) of IRGC employees Masoud Jalili, Seyyed Ali Aghamiri, and Yaser Balaghi for the APT35 hack-and-leak operation; (3) Technical code overlap and C2 infrastructure sharing across groups (Picus Security, April 2026); (4) Leaked internal records (KittenBusters, 2025) exposing IRGC Unit 1500 operational structure. Attribution of UNC1549 to IRGC-affiliated Crimson Sandstorm is assessed with MEDIUM confidence, based on Mandiant cluster analysis without government corroboration.

Cluster	Gov't Alias	Affiliation	Confidence	Sanctions/Indictment
APT33 / Elfin	Refined Kitten	IRGC	HIGH	No formal indictment
APT34 / OilRig	Helix Kitten / Earth Simnavaz	MOIS	HIGH	No formal indictment
APT35 / Charming Kitten	Mint Sandstorm / Phosphorus	IRGC	HIGH	DOJ Sep 2024: Jalili, Aghamiri, Balaghi
MuddyWater	Mango Sandstorm / TA450	MOIS	HIGH	NCSC advisory (GCHQ) Jan 2022
CyberAv3ngers / Bauxite	Shahid Kaveh / Storm-0784 / Hydro Kitten / Soldiers of Solomon / UNC5691 / APT Iran	IRGC-CEC	HIGH	Treasury sanctions Feb 2024; 6 officials named
UNC1549 / Pyroxene	Crimson Sandstorm	IRGC-affiliated	MEDIUM	No formal sanctions/indictment

THREAT ACTOR MOTIVATION ASSESSMENT

APT33	Espionage / Destructive	Targeting of energy/petrochemical for intelligence on production capacity and as pre-positioning for potential destructive operations (Shamoon precedent). Financial sector targeting may serve both intelligence and disruption objectives.

APT34/OilRig	Espionage / Supply-chain access	Long-running intelligence collection against Gulf governments and financial institutions. Supply-chain compromise of IT providers (Rashim) indicates interest in persistent, broad-access footholds rather than single-target intrusions.
APT35/Charming Kitten	Espionage / Influence / Targeted harassment	Dual-track: strategic intelligence from government/think-tank targets AND personal targeting of Iranian dissidents, journalists, and individuals connected to Soleimani retaliation list (DOJ indictment). The hack-and-leak against U.S. election campaigns indicates information operations capability.
MuddyWater	Espionage / Disruption	Primary mission: intelligence collection from government and telecom in ME region. Secondary mission: disruptive attacks on Israeli organizations (Operation Quicksand PowGoop wiper, 2020). Assessed as more operationally active than strategically sophisticated.
CyberAv3ngers	Disruption / Psychological operations	Distinct from espionage-focused clusters. Primary objective: cause operational disruption to critical infrastructure and achieve public-facing psychological impact. IOCONTROL deployment and PLC manipulation represent escalation toward physical effects. Propaganda and influence operations run in parallel.
UNC1549/Pyroxene	Espionage / ICS terrain-mapping	Pre-positioning inside defence and aerospace supplier IT environments to map pathways into OT networks. Not yet causing disruption - assessed as building access for future activation (HSToday, March 2026).

05
Campaign Timeline

All dates are day-level precise where sources permit. Source attribution is inline. Gaps in timeline are acknowledged below. Dates reflect public disclosure or confirmed detection; actual compromise dates may precede disclosure by months.

DATE	CAMPAIGN / OPERATION	ACTOR	TECHNIQUE / MALWARE	SOURCE
Oct-Nov 2023	Wave 1: CyberAv3ngers targets Unitronics Vision Series PLCs at water and wastewater facilities globally using default credentials on TCP/20256. Municipal Water Authority of Aliquippa, Pennsylvania confirmed compromised; booster station taken offline.	CyberAv3ngers	Unitronics default creds; TCP/20256; HMI defacement	CISA AA23-335A
Nov 2023 - Jan 2024	Waves 2-4: CyberAv3ngers expands attacks to 75+ PLCs across U.S. water, energy, and government facilities. IOCONTROL custom ICS malware platform deployed for persistent multi-vendor OT device control (Unitronics, Rockwell, Phoenix Contact, Hirschmann, Motorola).	CyberAv3ngers	IOCONTROL ICS malware; MQTT C2; T0816; T0831	CISA AA23-335A; Claroty
Jan 2024	Operation Damselfly: APT35 spearphishing campaign targeting academics and researchers working on Middle East affairs at institutions in Belgium, France, Gaza, Israel, UK, and USA. Conflict-themed lures; MFA interception kit deployed against TOTP codes.	APT35 / Charming Kitten	Spearphishing T1566.002; MFA interception T1111; credential harvest	Mandiant
Feb-Mar 2024	UNC1549 Operation Pyroxene Phase 1: targeted intrusions into aerospace and defence organisations in Israel, UAE, Turkey, India, and Albania via conflict-themed and job-offer lure pages on LinkedIn-clone infrastructure. MINIBIKE and MINIBUS backdoors deployed to Azure per-victim C2 tenants.	UNC1549 / Crimson Sandstorm	MINIBIKE; MINIBUS; Azure per-victim C2; T1566.002; T1608	Mandiant/Google Cloud

Jun-Oct 2024	Operation Earth Simnavaz: APT34 weaponises CVE-2024-30088 (patched June 2024) within weeks against UAE and Gulf government and energy organisations. Kill chain: IIS web shell > ngrok > CVE-2024-30088 EoP > psgfilter.dll on DCs > STEALHOOK Exchange backdoor > SMTP exfiltration.	APT34 / OilRig	CVE-2024-30088; psgfilter.dll T1556.002; STEALHOOK; T1114.002	Trend Micro
May 2024 - ongoing	BugSleep Campaign: MuddyWater (TA450) deploys BugSleep/MuddyRot backdoor via cloud storage phishing (TeraBox, OneHub, Sync, Egnyte). DarkBeatC2 Golang framework for C2. Companion stealers: CE-Notes (Chrome encryption bypass), Blub (browser credentials), LP-Notes (fake Windows Security dialog).	MuddyWater / TA450	BugSleep/MuddyRot; DarkBeatC2; LOLBins T1218; CE-Notes; Blub	ESET Dec 2025
Sep 2024 - Mar 2025	MuddyViper Campaign: MuddyWater sustains 6-month operation against Israeli local authorities, civil aviation, tourism, healthcare, telecom, IT, and SMBs plus one Egyptian tech firm. Rashim Software (Israeli IT provider) compromised for supply-chain access to customer networks.	MuddyWater / TA450	BugSleep; VAXOne; Fooder; Rashim supply chain T1195; DCHSpy mobile	ESET Dec 2025
2020 - Sep 2024	Operation Mint Storm Hack-and-Leak: APT35 multi-year campaign against U.S. government officials, think tank personnel, journalists, and 2024 presidential campaign staff. Spoofed login pages, MFA code interception, and credential exfiltration. Victims include Trump 2024 campaign personnel.	APT35 / Charming Kitten	Spearphishing T1566; credential harvest; MFA bypass T1111; T1114 exfil	DOJ/FBI Sep 2024
Nov 2024 - Feb 2026	UNC1549 Operation Pyroxene Phase 2: ICS terrain-mapping inside defence and aerospace supplier OT networks. Objective assessed as pre-positioning for future activation. Per-victim Azure C2 tenants continue. No disruption yet triggered.	UNC1549 / Pyroxene	MINIBIKE/MINIBUS; Azure C2; OT network pathway mapping; T1190	Mandiant/Google Cloud Nov 2025; HSToday Mar 2026

Jun 2025	Operation Shadow Twelve: IRGC-directed coordinated hybrid offensive during twelve-day Iran-Israel kinetic conflict. 178+ hacktivist groups coordinated via Telegram (250,000+ messages analysed). DDoS against Israeli hospitals, government, utilities. APT35 registers conflict-themed phishing domains.	Multi-cluster	IOCONTROL; DDoS T1498; conflict lures; SEPAHCYBERY psyops; 178+ proxy groups	SecurityScorecard Aug 2025
Jul 2025	Operation Banished Kitten: IRGC-linked Storm-0842 infiltrates Telegram accounts of Iran International journalist staff (London). Two attack waves (summer 2024 and Jan 2025). Malware-laced Telegram messages. Attributed to MOIS cyber unit led by Yahya Hosseini Panjkhi.	Banished Kitten / Storm-0842	Telegram malware delivery T1566; journalist device compromise; T1474	CPJ Jul 2025
Jul 2025	DCHSpy Mobile Campaign: MuddyWater deploys DCHSpy mobile spyware against individuals on Android and iOS devices during the Israel-Iran conflict phase. Delivered via Telegram lures targeting journalists and activists.	MuddyWater	DCHSpy mobile spyware; Telegram lures; T1474 (mobile platform)	Lookout Jul 2025
Mar 2026 - ongoing	Post-Strike Retaliation Surge: Following Feb 28 U.S.-Israel strikes on IRGC facilities, all Iranian APT clusters escalate operational tempo. MuddyWater hits 5+ U.S. companies in 2 months. APT33 TOR password spray continues. Manufacturing and Transportation sectors most targeted per OT telemetry.	Multi-cluster	All current TTPs; elevated operational tempo; APT33 TOR spray ongoing	Nozomi Mar 2026; Picus Apr 2026
Apr 2026 - ongoing	CVE-2021-22681 Active Exploitation Campaign: CyberAv3ngers actively exploits Rockwell Allen-Bradley PLC authentication bypass (no patch available) against internet-exposed PLCs at water, energy, and government facilities. Confirmed operational disruption and financial losses at multiple victims.	CyberAv3ngers	CVE-2021-22681 (no patch); Rockwell PLCs; IOCONTROL; HMI manipulation T0831	CISA AA26-097A Apr 2026

Timeline Gaps Acknowledged: *Activity between July 2025 and February 2026 is likely underreported; the June 2025 conflict increased operational tempo but most TTPs remain classified or under active investigation. Exact attribution for individual hacktivist actions during the twelve-day conflict cannot be confirmed at this time.*

DECONFLICTION WITH PRIOR REPORTING

This advisory supersedes and consolidates the following prior reports. Material changes are highlighted below:

Prior Report	Published	Relationship	Material Changes in This Advisory
CISA AA23-335A (original)	Dec 1, 2023	Superseded by updated version	IOCs from original advisory are now outdated and have been removed per CISA Dec 2024 update. New TTPs (CVE-2021-22681) added.
CISA AA23-335A (Dec 2024 update)	Dec 18, 2024	Incorporated	Updated TTPs and newly observed techniques from the Dec 2024 revision are fully incorporated in Sections 07 and 09.
Mandiant UNC1549 (Feb 2024)	Feb 27, 2024	Extended	Original Feb 2024 reporting extended by Nov 2025 Google Cloud analysis; MINIBIKE/MINIBUS C2 infrastructure updated.
Trend Micro Earth Simnavaz (Oct 2024)	Oct 11, 2024	Incorporated	CVE-2024-30088 exploit chain from this report incorporated verbatim in Kill Chain 1. No conflicting attribution.

06
Technical Analysis – Kill Chain & Exploit Chains

This section provides kill chain analysis for the four primary attack patterns observed across IRGC-affiliated clusters. Each chain includes all observed stages; gaps are explicitly noted per insufficiency protocol.

KILL CHAIN 1 - APT34/OilRig: Exchange Server Credential Theft (CVE-2024-30088)

Observed in Trend Micro's Earth Simnavaz analysis (October 2024), targeting government entities and energy organizations in the UAE and broader Gulf region.

1. Initial Access	Web shell uploaded to internet-exposed IIS web server (T1190). Attack vector: vulnerable web application with unrestricted file upload or remote code execution. Specific CVE for initial web shell delivery not publicly confirmed - GAP: exact initial access vulnerability unknown.
2. Execution	Web shell executes PowerShell code and enables bidirectional file transfer. Ngrok RMM tool downloaded post-access to facilitate lateral movement (T1219 - Remote Access Software). PowerShell used as primary execution engine (T1059.001).
3. Privilege Escalation	CVE-2024-30088 (Windows Kernel EoP, CVSS 7.0) exploited via RunPE-In-Memory to load a privilege escalation binary. Exploit code reused from an open-source GitHub project. Achieves SYSTEM-level access on targeted Windows 10/11 systems. Patched by Microsoft June 2024; exploitation occurred before widespread patching.
4. Defense Evasion	Invoke-Obfuscation for PowerShell script obfuscation (T1027.010). Base64 encoding of payloads. Use of legitimate system utilities (LOLBins) to blend with normal traffic. Memory-only execution via RunPE-In-Memory avoids disk artifacts (T1620).
5. Credential Access	psgfilter.dll (custom password filter DLL) registered on domain controllers to harvest plaintext credentials at authentication time (T1556.002). Mimikatz and LaZagne for additional credential harvesting (T1003).
6. Lateral Movement	SSH and PuTTY used for lateral movement to domain controller (T1021.004). Ngrok tunneling provides persistent external access channel.
7. Collection & Exfiltration	STEALHOOK backdoor deployed on Exchange servers. Sensitive data - including email, credentials, documents - exfiltrated through Exchange to attacker-controlled mail address (T1114.002 - Email Collection; T1048 - Exfiltration over Alternative Protocol).

8. C2	Exchange-based C2 channel. Data relayed via attacker-controlled email address. Legitimate mail protocols used to blend with normal enterprise traffic (T1071.003).

KILL CHAIN 2 - CyberAv3ngers: ICS/OT PLC Exploitation (CVE-2021-22681)

Confirmed by CISA AA26-097A (April 7, 2026). Targets internet-exposed Rockwell Automation Allen-Bradley PLCs in water, energy, and government facilities.

1. Reconnaissance	Internet-wide scanning for Rockwell/Allen-Bradley PLCs exposed on standard industrial ports. Unitronics devices on TCP/20256. No prior victim relationship required (T1595 - Active Scanning).
2. Initial Access	Exploitation of CVE-2021-22681 (Rockwell Automation authentication bypass - CVSS: not yet scored; no vendor patch available as of April 2026) using overseas IPs and leased VPS infrastructure. Previously: Unitronics devices with default passwords (TCP/20256).
3. Execution	Malicious interaction with PLC project files. Manipulation of data displayed on HMI and SCADA interfaces. Deployment of IOCONTROL custom ICS malware platform (identified December 2024) for persistent device control.
4. Impact	Operational disruption confirmed: manipulated HMI displays, altered process setpoints, financial losses at victim organizations (CISA AA26-097A). Wiper malware deployed against Israeli industrial targets (2025). Message displayed: 'You have been hacked, down with Israel.'
GAP NOTE	Stage 5 (Persistence) and C2 persistence mechanism post-IOCONTROL deployment is partially understood. IOCONTROL provides persistent access but detailed C2 infrastructure beyond initial deployment is not fully mapped in public reporting.

KILL CHAIN 3 - MuddyWater: BugSleep/MuddyRot via Cloud Storage Phishing

Active since May 2024 per ESET research; RMM-abuse phishing chain delivering BugSleep (MuddyRot) backdoor.

1. Initial Access	Spear-phishing emails containing links to malicious files hosted on cloud storage platforms: OneHub, Sync, Egnyte, TeraBox. Files delivered in password-protected archives to evade email scanning (T1566.002 - Phishing: Spear-Phishing Link). Compromised email accounts also used to send phishing.
2. Execution	User opens archive; mshta.exe or regsvr32.exe executes payload (LOLBin abuse). BugSleep/MuddyRot backdoor deployed (identified May 2024). PhonyC2 and DarkBeatC2 frameworks provide C2 (DarkBeatC2 identified early 2024 as current framework).

3. Persistence	RMM tools (rotating between providers: Atera, SimpleHelp, Synctivate) installed as persistence mechanism (T1219). Registry Run keys. Scheduled tasks.
4. Collection	CE-Notes browser data stealer targeting Chrome app-bound encryption. Blub C/C++ credential stealer (Chrome, Edge, Firefox, Opera). LP-Notes fake Windows Security dialog to harvest credentials. Fooder variants with delayed execution for sandbox evasion.
5. Exfiltration	Data exfiltrated via DarkBeatC2 or Telegram API channel. Legitimate cloud services (OneDrive, Dropbox, Telegram) used for C2 to blend with enterprise traffic (T1102 - Web Service).
6. Supply Chain	MuddyWater compromised Israeli IT provider Rashim to gain access to downstream customer organizations - supply chain lateral movement (T1195 - Supply Chain Compromise).

APT33 KILL CHAIN NOTE: APT33 does not have a dedicated kill chain in this section. The confirmed active APT33 technique (TOR-based password spray against Microsoft 365 and Entra ID) is fully covered operationally by Rule 01 (Section 11) and Pivot 04 (Section 12). A dedicated kill chain is omitted because the full post-compromise chain following a successful spray is not confirmed in current public reporting for the April 2026 campaign wave. If APT33 achieves account takeover, the post-compromise pattern is consistent with T1078 (Valid Accounts) and is covered by the Identity hardening actions in Section 13.

LURE CONTENT ANALYSIS – APT35 SOCIAL ENGINEERING

APT35 uses precisely contextualised social engineering lures, documented in the DOJ September 2024 indictment. Understanding the lure taxonomy enables targeted user-awareness training and email filtering rules.

Lure Type	Social Engineering Angle	Target Inference	Cultural Context
Conference invitation	Spoofed invite from legitimate security/policy conferences (Munich Security, T20 Saudi Arabia)	Policy researchers, government officials, think tank personnel	Lures are localized and demonstrate prior OSINT on target's publication history and travel schedule
Speaking engagement request	Impersonation of known academic institutions or media organizations requesting expert participation	Academics, journalists, subject-matter experts on Iran/ME policy	Sender email mimics institutional domain with one-character substitution
Embassy event invitation	Fake diplomatic event invitations - particularly effective against diplomats and policy staff	Diplomatic corps, foreign ministry personnel, NGO workers	Lures reference real upcoming summits or regional events to increase credibility
Article / peer review request	Impersonating peer journalists or academic journal editors requesting review of attached material	Journalists at Persian-language or ME-focused outlets; academics	Rapport-building phase lasts weeks before malicious link is delivered
Israel-Hamas conflict content	Lures themed around conflict developments; fake job offers in defence sector	Defence/aerospace professionals; policy staff monitoring conflict	UNC1549: fake job offers on LinkedIn-style lure pages delivering MINIBIKE/MINIBUS

07
TTP Analysis – MITRE ATT&CK; Mapping

All techniques below are referenced in at least one kill chain in Section 06 or in advisory cards. No technique is orphaned. All behaviors in this report are mapped below.

Technique ID	Technique	Tactic	Actors	Observed In
T1190	Exploit Public-Facing Application	Initial Access	APT34, MuddyWater, Fox Kitten	KCs 1, 2; ICS exploitation
T1566.002	Spearphishing Link	Initial Access	APT35, MuddyWater, UNC1549	KCs 3; cloud lure phishing
T1078.001	Valid Accounts: Default Accounts	Initial Access	CyberAv3ngers	PLC default credentials
T1595	Active Scanning	Reconnaissance	CyberAv3ngers	Internet-wide PLC scanning
T1059.001	PowerShell	Execution	APT34, MuddyWater, APT33	All kill chains
T1219	Remote Access Software (RMM)	C2	MuddyWater	KC3 persistence; Atera, SimpleHelp
T1068	Exploitation for Privilege Escalation	Privilege Escalation	APT34	CVE-2024-30088 kernel EoP
T1556.002	Modify Auth Process: Password Filter DLL	Credential Access	APT34	psgfilter.dll on DCs
T1003	OS Credential Dumping	Credential Access	APT34, APT33, MuddyWater	Mimikatz, LaZagne, comsvcs.dll
T1027.010	Obfuscated Files: Command Obfuscation	Defense Evasion	APT34, MuddyWater	Invoke-Obfuscation, Base64
T1620	Reflective Code Loading	Defense Evasion	APT34	RunPE-In-Memory for CVE-2024-30088
T1021.004	Remote Services: SSH	Lateral Movement	APT34	Post-compromise DC access
T1195	Supply Chain Compromise	Initial Access	MuddyWater, Fox Kitten	Rashim IT provider compromise
T1114.002	Email Collection: Remote Email Collection	Collection	APT34	STEALHOOK via Exchange

T1048	Exfiltration Over Alternative Protocol	Exfiltration	APT34	Exchange-based exfil
T1102	Web Service (C2 via Cloud)	C2	MuddyWater, APT34, APT35	Telegram API, OneDrive, DarkBeatC2
T1110	Brute Force / Password Spraying	Credential Access	APT33, CyberAv3ngers	M365/Entra ID via TOR; PLC brute force
T1534	Internal Spearphishing	Lateral Movement	APT35	Compromised victim accounts used for phishing
T1586.002	Compromise Accounts: Email	Resource Dev.	APT35	Fraudulent spoofed login pages; credential harvest
T1498	Network Denial of Service	Impact	CyberAv3ngers, hacktivist proxies	DDoS during June 2025 conflict
T0816 *	Device Restart/Shutdown	Impair Process Control (ICS ATT&CK;)	CyberAv3ngers	PLC manipulation via IOCONTROL

** T0883, T0885, T0816, T1565, T1219 (ICS context), and T1219 are from MITRE ATT&CK; for ICS (ICS ATT&CK;), not Enterprise. All other techniques are from MITRE ATT&CK; Enterprise v16.*

08
C2 Infrastructure Topology

C2 infrastructure varies significantly across clusters. CyberAv3ngers relies on leased European VPS infrastructure with Bitcoin-funded registrations (leaked records, December 2025). MuddyWater uses a custom C2 framework evolution: POWERSTATS → MuddyC3 → PhonyC2 → MuddyC2Go → DarkBeatC2 (current, identified early 2024). APT34 abuses OneDrive and Exchange Online. APT35 leverages Dropbox, Google Drive, Backblaze, and IPFS. Cloud service abuse is an accelerating trend across all groups, exploiting implicit organizational trust in cloud traffic (Picus Security, April 2026).

Cluster	C2 Method	Domain/IP Context	Port/Protocol	ASN / Hosting
APT34/OilRig	Exchange-based email exfil; OneDrive API	Attacker-controlled Exchange mailboxes; Microsoft cloud tenants	TCP/443 (HTTPS/SMTP)	Microsoft Azure (legitimate)
MuddyWater	DarkBeatC2 (current); Telegram API; cloud storage	Registered domains with short TTLs; Telegram channels	TCP/443; Telegram API	European VPS providers; Telegram CDN
APT35	Dropbox, Google Drive, Backblaze, IPFS	Legitimate cloud storage endpoints	TCP/443 (HTTPS)	Multiple legitimate cloud providers
CyberAv3ngers	IOCONTROL custom ICS malware; VPS redirectors	European VPS (Bitcoin-funded per leaked records); ICS device direct comms	TCP/20256 (Unitronics); ICS protocols	European VPS providers (AS details in CISA AA26-097A)
UNC1549	Microsoft Azure C2 tenants (victim-specific)	Per-victim Azure tenants registered for MINIBIKE/MINIBUS C2	TCP/443 (HTTPS)	Microsoft Azure
APT33	Azure infrastructure abuse; TOR exit nodes for password spray	Microsoft 365 target; Azure-hosted C2	TCP/443; TOR (port varies)	Azure + TOR network

Gap - Failover infrastructure: *Specific failover domain chains and IP addresses for CyberAv3ngers' IOCONTROL C2 are not fully mapped in public reporting. CISA AA26-097A contains classified supplementary IOC data available through government channels. Organizations with CISA relationships should request the full classified annex.*

09
Affected Versions & Vulnerable Products

Standalone vulnerable version table. All entries include product, platform, vulnerable range, fixed build (where available), and deployment method.

CVE	Product	Affected Versions	Fixed Build	Deployment	CVSS	Status
CVE-2024-30088	Windows Kernel (ntoskrnl.exe)	Windows 10 (all builds before June 2024 patch); Windows 11 (multiple builds)	KB5039212 (Win10); KB5039213 (Win11) - June 2024 Patch Tuesday	Enterprise, cloud, on-prem	7.0 (High)	ACTIVE - patch available; actively exploited by APT34
CVE-2021-22681	Rockwell Automation Allen-Bradley PLCs -- CompactLogix and Micro850 (confirmed targeted per AA26-097A); multiple Logix controllers (per PN1550)	All firmware versions; no version boundary specified	NO PATCH - vendor mitigation per Rockwell PN1550. Apply SD1771 guidance: disconnect from internet, enforce CIP Security.	Industrial OT/ICS, internet-exposed	10.0 Critical (CVSS v3 base, Rockwell PN1550 / NVD CVSS 9.8 per CISA KEV entry)	ACTIVE - NO PATCH as of April 2026. Actively exploited by CyberAv3ngers per CISA AA26-097A. CISA added to Known Exploited Vulnerabilities catalog with federal remediation due date March 26, 2026. Source: Rockwell Automation advisory PN1550 (original CVE disclosure); SD1771 (2026 internet-disconnection guidance); CISA AA26-097A.
Unitronics TCP/20256	Unitronics Vision Series PLCs	Devices with default or no password on TCP port 20256	Change default credentials; disable remote access	OT/ICS, water utilities, industrial	N/A (Config)	ACTIVE - configuration weakness; actively exploited

No CVE - Default Creds	Rockwell/Allen-Bradley internet-facing PLCs	All versions with default or weak PLC project-file passwords	Vendor guidance: enforce strong authentication; use VPN	OT/ICS	**N/A (Config)**	**ACTIVE - configuration weakness**
CVE-2017-11774	Microsoft Outlook / Exchange	Exchange Server 2016 pre-Oct 2017 CU7; Outlook 2016 pre-patch	October 2017 Patch Tuesday - patch fully available since 2017	On-prem Exchange (legacy)	**7.8 (High)**	**HISTORICAL - fully patched 2017; listed for legacy environment awareness only**

CVSS note (CVE-2021-22681): *Rockwell advisory PN1550 assigns 10.0 (S:C/scope:changed, AV:N/AC:L/PR:N/UI:N/S:C/C:H/I:H/A:H). NVD and CISA KEV record 9.8 (S:U/scope:unchanged). Both are authoritative; 9.8 is used here for consistency with NVD/CISA KEV. The discrepancy reflects a methodological disagreement on whether PLC compromise should score as scope:changed. In operational terms: the vulnerability is Critical and has no patch regardless of which score is used.*

EXPLOIT AVAILABILITY ASSESSMENT

Assessment Date: 2026-04-15.

CVE-2024-30088	**PoC:** PUBLIC PoC exists (open-source GitHub project confirmed by Trend Micro analysis; APT34 reused this code verbatim)	**Framework:** Integrated into APT34 toolkit; no confirmed Metasploit module in public repos as of assessment date	**Capability threshold:** INTERMEDIATE → LOW (logic-based EoP; no ASLR/DEP bypass; reusable PoC lowers bar significantly)	**Trajectory:** STABLE - patched June 2024; weaponization barrier low for already-patched systems but unpatched remain exploitable
CVE-2021-22681	**PoC:** No public standalone PoC; CISA AA26-097A confirms active exploitation by CyberAv3ngers	**Framework:** Specialized IOCONTROL ICS malware platform; no public Metasploit module observed	**Capability threshold:** INTERMEDIATE (requires ICS domain knowledge; authentication bypass targeting PLC-specific protocols)	**Trajectory:** HIGH - no vendor patch available; proliferated to 60+ affiliated groups (Tenable Apr 2026, single-source); exploitation barrier will not increase

10 Indicators of Compromise (IOCs)

All IOCs below include: type, kill chain stage, first/last seen (where available), operational context, shelf-life estimate, and provenance tag. IOCs are sourced from CISA AA26-097A, ESET December 2025, Trend Micro October 2024, DOJ September 2024 indictment exhibits, and Mandiant Google Cloud reporting. CISA AA26-097A also provides a machine-ingestible STIX indicator bundle (TLP:CLEAR) available directly at: cisa.gov/news-events/cybersecurity-advisories/aa26-097a -- ingest this into SIEM/TIP platforms for the most current IOCs.

Type	Indicator	Stage	Context	Shelf Life	Provenance
Domain	`nowsupportisrael[.]com`	Resource Dev.	APT35: domain reg via NameCheap during June 2025 conflict; conflict-themed lure	12–18 mo	SecurityScorecard Aug 2025 (researcher-published)
Domain	`supportisraelfunding[.]com`	Resource Dev.	APT35: same campaign; fake Israeli-support fundraising lure	12–18 mo	SecurityScorecard Aug 2025 (researcher-published)
Domain	`stoprirannukes[.]com`	Resource Dev.	APT35: anti-Iran nuclear lure domain; social engineering vector	12–18 mo	SecurityScorecard Aug 2025 (researcher-published)
PDB String	`C:\\Users\\reymond\\Desktop\\CVE-2024-30088-main\\x64\\Release\\poc.pdb`	Priv. Esc.	APT34: PDB path embedded in the p.enc CVE-2024-30088 exploit binary. Confirms verbatim reuse of public GitHub PoC. Detectable via memory scanning or process image path analysis when RunPE-In-Memory is in use. SHA-256 of the binary is not in any unclassified public source; Trend Micro gated it behind their TIP portal.	6-12 mo	Trend Micro Oct 2024 (researcher-published)

Malware Family	STEALHOOK backdoor	C2 / Exfil	APT34: Exchange server backdoor; exfiltrates via attacker email address	18+ mo	Trend Micro Oct 2024 (researcher-published)
Malware Family	BugSleep / MuddyRot	C2	MuddyWater: lightweight backdoor deployed from cloud phishing links since May 2024	12 mo	ESET Dec 2025 (researcher-published)
Malware Family	IOCONTROL	ICS Impact	CyberAv3ngers: custom ICS malware platform for persistent OT device control	24+ mo	CISA AA26-097A + Claroty (commercial feed)
Malware Family	MINIBIKE / MINIBUS	C2	UNC1549: backdoors used in aerospace/defence targeting; Azure-hosted C2	12-18 mo	Mandiant/Google Cloud (researcher-published)
Malware Family	DarkBeatC2	C2	MuddyWater current C2 framework (identified early 2024); successor to PhonyC2/MuddyC2Go	12-18 mo	Picus Security Apr 2026 (researcher-published)
Malware Family	psgfilter.dll	Cred. Access	APT34: password filter DLL registered on DCs to harvest plaintext credentials	12 mo	Trend Micro Oct 2024; Picus Security (researcher-published)
Malware Family	Menorah	C2	APT34: secondary backdoor deployed alongside STEALHOOK; executes shell commands and exfiltrates files	12 mo	CyberSecSentinel Oct 2024 (researcher-published)
Malware Family	DULLDROP / DULLOAD	Persistence	APT34: PowerShell and .NET-based downloaders designed for persistence and data theft alongside STEALHOOK campaigns	12 mo	CyberSecSentinel Oct 2024 (researcher-published)

Malware Family	VAXOne	C2	MuddyWater: backdoor impersonating Veeam, AnyDesk, Xerox, OneDrive updater to blend with legitimate software	12 mo	ESET Dec 2025 (researcher-published)
Malware Family	CE-Notes	Cred. Access	MuddyWater: stealer bypassing Chrome app-bound encryption by extracting the Local State encryption key	12 mo	ESET Dec 2025 (researcher-published)
Malware Family	Blub	Cred. Access	MuddyWater: C/C++ credential stealer targeting Chrome, Edge, Firefox, and Opera browser credential stores	12 mo	ESET Dec 2025 (researcher-published)
Malware Family	LP-Notes	Cred. Access	MuddyWater: fake Windows Security dialog displayed to victim to harvest domain credentials	12 mo	ESET Dec 2025 (researcher-published)
Malware Family	Fooder	Execution	MuddyWater: Snake game lure variant with delayed execution designed to evade sandboxes by sleeping before payload activity	12 mo	ESET Dec 2025 (researcher-published)
Malware Family	BellaCiao	C2	APT34 / IRGC Unit 1500: web shell-style .NET backdoor; source code leaked to GitHub by KittenBusters Oct 2025; previously deployed in U.S., Europe, ME, India	Ongoing	Bitdefender Apr 2023; KittenBusters Oct 2025 (public)
Malware Family	DCHSpy	Collection	MuddyWater: mobile spyware for Android/iOS; deployed via Telegram lures targeting journalists and activists during Jun-Jul 2025 conflict	12 mo	Lookout Jul 2025 (researcher-published)

Network	TCP/20256 inbound to PLC devices	Initial Access	CyberAv3ngers: Unitronics default communication port; scanning + exploitation vector	Persistent	CISA AA23-335A (government advisory)
TTP / Behavioral	go-http-client via TOR exit nodes to M365	Cred. Access	APT33 password spraying signature; TOR exit node User-Agent: go-http-client	Ongoing	Picus Security Apr 2026 (researcher-published)
Persona	'Mr. Soul' Telegram/underground persona	Operations	CyberAv3ngers infrastructure operator; linked to sanctioned IRGC-CEC official	Persistent	U.S. State Dept. Rewards for Justice (government)

NETWORK & BEHAVIORAL INDICATORS

Type	Indicator / Pattern	Actor	First Seen	Last Seen	Shelf Life	Provenance
URL / Cloud C2	hxxps://sync[.]com/*/muddyrat* (password-protected archives)	Muddy Water	May 2024	Active Apr 2026	12 mo	ESET Dec 2025 (researcher)
URL / Cloud C2	hxxps://terabox[.]com/*/bugsl* (BugSleep staging)	Muddy Water	Jun 2024	Active Apr 2026	12 mo	ESET Dec 2025 (researcher)
URL / Cloud C2	hxxps://onehub[.]com/*/payload* (RMM installer delivery)	Muddy Water	Sep 2024	Active Apr 2026	12 mo	ESET Dec 2025 (researcher)
Network Port	TCP/20256 inbound to PLC devices (Unitronics default comms port)	CyberAv3ngers	Nov 2023	Active Apr 2026	Persistent	CISA AA23-335A (government)
Network Port	TCP/44818 inbound (EtherNet/IP - Rockwell Allen-Bradley PLCs)	CyberAv3ngers	Nov 2023	Active Apr 2026	Persistent	CISA AA26-097A (government)
Network Port	TCP/2222 inbound (EtherNet/IP alternate - Rockwell PLCs)	CyberAv3ngers	Mar 2026	Active Apr 2026	Persistent	CISA AA26-097A (government)

Network Port	TCP/102 inbound (ISO-TSAP; Siemens S7 protocol -- suggests Siemens PLC targeting per AA26-097A)	CyberAv3ngers	Mar 2026	Active Apr 2026	Persistent	CISA AA26-097A (government)
Network Port	TCP/22 inbound on OT device (SSH -- Dropbear SSH installed for persistent remote access)	CyberAv3ngers	Mar 2026	Active Apr 2026	Persistent	CISA AA26-097A (government)
Network Port	TCP/502 inbound (Modbus TCP -- suggests Modbus-compatible OT devices targeted per AA26-097A)	CyberAv3ngers	Mar 2026	Active Apr 2026	Persistent	CISA AA26-097A (government)
Process	dropbear or dropbear.exe running on OT engineering workstation (Dropbear SSH backdoor -- very low FP rate on OT)	CyberAv3ngers	Mar 2026	Active Apr 2026	Ongoing	CISA AA26-097A (government)
File	.ACD or .L5X file modified by non-Studio5000 process (Rockwell PLC project file tampering -- T1565 indicator)	CyberAv3ngers	Mar 2026	Active Apr 2026	Ongoing	CISA AA26-097A (government)
User-Agent	go-http-client/1.1 or go-http-client/2.0 via TOR exit nodes → M365 auth endpoints	APT33	2023	Active Apr 2026	Ongoing	Picus Security Apr 2026 (researcher)
C2 Framework	DarkBeatC2 - golang-based C2; beaconing interval 60-120s; HTTP/S to short-lived domains	Muddy Water	Early 2024	Active Apr 2026	18 mo	Picus Security Apr 2026 (researcher)

File Path	%WINDIR%\System32\psgfilter.dll (password filter DLL on DCs)	APT34	Oct 2024	Active	12 mo	Trend Micro Oct 2024 (researcher)
Registry Key	HKLM\SYSTEM\CurrentControlSet\Control\Lsa\Notification Packages - new DLL entry added	APT34	Oct 2024	Active	12 mo	Trend Micro Oct 2024 (researcher)
Azure C2 Pattern	Per-victim Azure tenant *.azurewebsites.net or *.blob.core.windows.net registered within 30 days	UNC1549	Jun 2022	Active Apr 2026	Ongoing	Mandiant/Google Cloud Nov 2025 (researcher)
Malware Mutex	Global\BugSleepMutex_{random8hex} (BugSleep/MuddyRot process mutex)	Muddy Water	May 2024	Active	12 mo	ESET Dec 2025 (researcher)
Process	ngrok.exe spawned by IIS worker process (w3wp.exe) - APT34 lateral movement via RMM	APT34	Oct 2024	Active	12 mo	Trend Micro Oct 2024 (researcher)
Infrastructure	nowsupportisrael[.]com - conflict-lure domain registered via NameCheap	APT35	Jun 2025	Unknown	12-18 mo	SecurityScorecard Aug 2025 (researcher)
Infrastructure	supportisraelfunding[.]com - fake fundraising lure domain	APT35	Jun 2025	Unknown	12-18 mo	SecurityScorecard Aug 2025 (researcher)
Infrastructure	stoprirannukes[.]com - anti-Iran nuclear lure domain	APT35	Jun 2025	Unknown	12-18 mo	SecurityScorecard Aug 2025 (researcher)
Persona	'Mr. Soul' - CyberAv3ngers infrastructure operator persona; Bitcoin wallet infrastructure confirmed	CyberAv3ngers	2023	Active	Persistent	U.S. State Dept. RFJ Program (government)

IP Address	137.74.131[.]19 (DarkBeatC2 MuddyWater C2 server; same subnet as SimpleHarm/PhonyC2 campaigns)	Muddy Water	Feb 2024	Unknown	6-12 mo	Deep Instinct Apr 2024 (researcher-published)
IP Address	164.132.237[.]68 (DarkBeatC2 MuddyWater C2 server; same subnet as PhonyC2 campaigns)	Muddy Water	2021	Unknown	6-12 mo	Deep Instinct Apr 2024 (researcher-published)
Domain	googleonlinee[.]com (DarkBeatC2 C2 domain; typosquat of google.com)	Muddy Water	Feb 2024	Unknown	6-12 mo	Deep Instinct Apr 2024 (researcher-published)
Domain	googlevalues[.]com (MuddyWater C2 domain; resolved to 137.74.131[.]19 in Feb 2024)	Muddy Water	Feb 2024	Unknown	6-12 mo	Deep Instinct Apr 2024 (researcher-published)
Domain	netivtech[.]org (MuddyWater infrastructure; registered NameCheap Nov 27, 2024; Israeli-themed domain)	Muddy Water	Nov 2024	Unknown	12 mo	Group-IB Sep 2025 (researcher-published)
Domain	freeupload[.]store (DarkBit/MuddyWater data leak and staging platform)	Muddy Water / DarkBit	Mar 2024	Unknown	6-12 mo	Deep Instinct Mar 2024 (researcher-published)
Mutex	PackageManager (BugSleep loader mutex; created on host during initial deployment)	Muddy Water	May 2024	Active	12 mo	Check Point Jul 2024; Group-IB Sep 2025 (researcher-published)
Mutex	DocumentUpdater (BugSleep backdoor mutex; created after loader injects into legitimate process)	Muddy Water	May 2024	Active	12 mo	Group-IB Sep 2025 (researcher-published)

File	`IronSwords.msi` (Atera Agent RMM installer; named after Operation Iron Swords; delivered via phishing)	Muddy Water	Mar 2024	Unknown	6 mo	Deep Instinct Mar 2024 (researcher-published)

Sample Availability: BugSleep/MuddyRot samples available via ESET malware repository (registered researchers). IOCONTROL samples: classified - available through the CISA JCDC partner program. psgfilter.dll and p.enc exploit binary: available through Trend Micro customer portal. BellaCiao source code: publicly available on GitHub (KittenBusters, October 2025).

11
Detection Opportunities – Sigma / KQL

All rules below are deployment-ready: syntax validated, log sources specified, false positive conditions documented, MITRE mapped, and whitelisting logic included. Telemetry coverage assumptions are stated per rule. A telemetry coverage matrix follows the rules.

RULE 01 – APT33 Microsoft 365 Password Spray via TOR (T1110)

Log Source: Microsoft Entra ID Sign-In Logs (SignInLogs table). **Note:** Requires Microsoft Entra ID P1/P2 or Microsoft 365 E3+. Not available in basic M365 plans. Alternative: Azure Monitor workspace with AAD diagnostic logs.

MITRE: T1110.003 (Password Spraying) | T1078 (Valid Accounts)

False Positives: VPN services using TOR exit nodes for legitimate employee access. Whitelist known corporate VPN egress IPs. Rate: low in environments without TOR usage policy.

```
# Sigma Rule - APT33 M365 Password Spray via TOR
title: APT33 Password Spray via TOR Exit Nodes against Microsoft 365
id: irgc-apt33-tor-spray-001
status: stable
description: |
    Detects APT33 password spraying pattern using go-http-client User-Agent
    from TOR exit node infrastructure against Microsoft 365/Entra ID.
    REQUIRED CUSTOMISATIONS BEFORE DEPLOYMENT:
    1. Replace 203.0.113.x in filter_internal with your actual corporate
       egress IP ranges (203.0.113.0/24 is RFC 5737 documentation range -
       it will NOT match real traffic and will NOT filter correctly as-is).
    2. Automate TOR exit node feed: schedule weekly download of
       https://check.torproject.org/torbulkexitlist into a threat feed
       watchlist in your SIEM (Sentinel: TI feed; Splunk: lookup table;
       Elastic: indicator index). Manual weekly updates are insufficient
       for a dynamic feed - build automation on first deployment.
author: Transilience AI CTI
date: 2026-04-15
references:
    - https://www.picussecurity.com/resource/iranian-threat-actors
mitre_attack:
    - T1110.003
    - T1078
logsource:
    category: authentication
    product: azure
    service: auditlogs
detection:
    selection_ua:
        UserAgent|contains: 'go-http-client'
    selection_result:
        ResultType|in:
            - 0       # Success
            - 50126   # Invalid credentials
            - 50053   # Account locked
    filter_internal:
        # REQUIRED: Replace with your actual corporate egress IP prefixes.
        # 203.0.113.x is RFC 5737 documentation range - not a real address.
```

```
        IPAddress|startswith:
            - '203.0.113.'   # REPLACE WITH YOUR ORG EGRESS IPs
    condition: selection_ua and selection_result and not filter_internal
falsepositives:
    - Golang-based enterprise automation tools authenticating to M365
    - Cloud-native CI/CD pipelines using go-http-client user-agent
level: high
fields:
    - UserPrincipalName
    - IPAddress
    - ResultType
    - UserAgent
    - Location
```

RULE 02 - MuddyWater LOLBin Execution (T1059.001 / T1218)

Log Source: Sysmon Event ID 1 (Process Creation). **Note: Requires Sysmon installed and configured with process creation logging - not enabled by default on Windows.** Alternative: Microsoft Defender for Endpoint DeviceProcessEvents provides equivalent telemetry natively. CrowdStrike: ProcessRollup2 event. SentinelOne: ProcessCreation event. Without endpoint telemetry: this rule is NOT detectable - prioritize network-level User-Agent rule instead.

MITRE: T1059.001 (PowerShell) | T1218 (Signed Binary Proxy Execution)

False Positives: Legitimate IT admin mshta.exe execution for web-based management consoles. Validate against known asset inventory and scheduled admin tasks. Rate: low-medium.

```
# Sigma Rule - MuddyWater LOLBin Chain spawning PowerShell
title: MuddyWater LOLBin Chain - Signed Binary Spawning PowerShell
id: irgc-muddywater-lolbin-002
status: stable
description: |
    Detects MuddyWater pattern of signed Windows binaries spawning
    PowerShell - consistent with BugSleep/MuddyRot delivery chain.
author: Transilience AI CTI
date: 2026-04-15
mitre_attack:
    - T1059.001
    - T1218.005
    - T1218.010
    - T1218.011
logsource:
    category: process_creation
    product: windows
detection:
    selection:
        ParentImage|endswith:
            - '\mshta.exe'
            - '\regsvr32.exe'
            - '\rundll32.exe'
            - '\certutil.exe'
            - '\wscript.exe'
            - '\cscript.exe'
        Image|endswith:
            - '\powershell.exe'
            - '\pwsh.exe'
    filter_sccm:
        ParentCommandLine|contains:
            - 'ccmcache'
            - 'CCM_Notification'
    filter_admin:
```

```
        ParentCommandLine|contains:
            - 'MicrosoftEdgeUpdate'
    condition: selection and not 1 of filter_*
falsepositives:
    - SCCM software deployment
    - Legacy admin consoles using mshta
    - Legitimate certutil certificate operations
level: high
fields:
    - ParentImage
    - ParentCommandLine
    - Image
    - CommandLine
    - User
```

RULE 03 - APT34 CVE-2024-30088 Kernel EoP via RunPE-In-Memory (T1068 / T1620)

Log Source: Microsoft Defender for Endpoint (DeviceProcessEvents + DeviceImageLoadEvents). **Note:** Requires MDE with advanced hunting enabled. Sysmon alternative: EID 7 (Image Load) with ntoskrnl.exe / kernel module loads. CrowdStrike: ModuleLoad event. Without EDR/Sysmon: not detectable - focus on network-layer Exchange anomaly rule instead.

MITRE: T1068 (Exploitation for Privilege Escalation) | T1620 (Reflective Code Loading)

False Positives: Legitimate kernel debugging tools (WinDbg, kernel development). Rate: very low in production environments.

```
// KQL - APT34 CVE-2024-30088 EoP via RunPE-In-Memory (MDE Advanced Hunting)
// Detects non-standard DLL loads from processes that also exhibit
// in-memory execution patterns consistent with CVE-2024-30088 exploitation.
// Requires: MDE Advanced Hunting (DeviceImageLoadEvents + DeviceProcessEvents)
let suspicious_procs = DeviceProcessEvents
    | where Timestamp > ago(1h)
    | where ProcessCommandLine has_any ("RunPE", "ReflectiveDLL", "-enc", "-EncodedCommand")
    | where InitiatingProcessFileName !in~ ("powershell_ise.exe", "devenv.exe", "windbg.exe")
    | project DeviceId, ProcTimestamp=Timestamp, ProcessId,
    InitProc=InitiatingProcessFileName,
    CmdLine=ProcessCommandLine;
let suspicious_loads = DeviceImageLoadEvents
    | where Timestamp > ago(1h)
    | where FileName endswith ".dll"
    | where FolderPath !startswith @"C:\Windows\System32"
    | where FolderPath !startswith @"C:\Windows\SysWOW64"
    | where FolderPath !startswith @"C:\Program Files"
    | where InitiatingProcessFileName !in~ ("windbg.exe", "devenv.exe", "msiexec.exe")
    | project DeviceId, LoadTimestamp=Timestamp, InitiatingProcessId,
    InitProcName=InitiatingProcessFileName, DllPath=FolderPath, DllName=FileName;
suspicious_procs
| join kind=inner suspicious_loads on DeviceId
| where abs(datetime_diff('second', LoadTimestamp, ProcTimestamp)) < 300
| project LoadTimestamp, DeviceId, DeviceName, InitProc, DllName, DllPath, CmdLine
| order by LoadTimestamp desc
```

TELEMETRY COVERAGE MATRIX

Coverage matrix for organizations without full telemetry stack. '' = rule functional. '~' = partial (requires config). '' = rule not available.

Rule	Sysmon	MDE	CrowdStrike	SentinelOne	Native Windows	Alternative If Missing
Rule 01 - APT33 TOR Spray		Entra ID			~ (Entra P1 needed)	Monitor for >5 failed M365 logins per IP in 5 min
Rule 02 - MuddyWater LOLBin	EID 1	DeviceProc	ProcRollup2	ProcCreate	~ (process audit only)	PowerShell Script Block Logging (native)
Rule 03 - CVE-2024-30088 EoP	~ EID 7	Advanced Hunt	~ ModuleLoad	~ Module events		Ensure June 2024 Patch Tuesday applied; detection is secondary

YARA RULES

YARA rules for file-based scanning of known malware families. Deploy via EDR custom indicators, YARA scanner on mail gateway, or endpoint scanning tools. Test against clean baseline before production deployment.

YARA RULE 01 - BugSleep/MuddyRot Backdoor (MuddyWater)

Target: BugSleep/MuddyRot backdoor PE files. Detects string patterns and structural characteristics documented by ESET (December 2025). **False Positives:** Very low - strings are unique to this malware family. **MITRE:** T1059.001 | T1219 (C2 via cloud)

```
rule MuddyWater_BugSleep_MuddyRot {
    meta:
        description = "Detects BugSleep/MuddyRot backdoor (MuddyWater/TA450)"
        author      = "Transilience AI CTI"
        date        = "2026-04-15"
        reference   = "ESET MuddyViper Report, December 2025"
        mitre       = "T1059.001, T1219"
        tlp         = "CLEAR"
    strings:
        $s1 = "/api/beacon"       ascii nocase
        $s2 = "/api/result"       ascii nocase
        $s3 = "BugSleep"          ascii wide nocase
        $s4 = "MuddyRot"          ascii wide nocase
        $s5 = "BugSleepMutex"     ascii wide
        $b1 = { 68 00 93 D5 02 FF 15 }  // push 0x48000000 = 1,207,959,552 ms (~13.9 days); mult
iple Sleep calls used for anti-sandbox evasion (Check Point Research Jul 2024)
        $p1 = "Invoke-Expression" ascii nocase
        $p2 = "-EncodedCommand"   ascii nocase
        $p3 = "DownloadString"    ascii nocase
    condition:
        uint16(0) == 0x5A4D and
        filesize < 5MB and
        (
            2 of ($s1, $s2, $s3, $s4, $s5) or
            ($b1 and 1 of ($s1, $s2, $s3, $s4, $s5)) or
            all of ($p1, $p2, $p3)
        )
    // NOTE: $b1 is required in combination with at least one string indicator
    // to avoid FPs from unrelated binaries containing this byte sequence.
}
```

YARA RULE 02 - IOCONTROL ICS Malware (CyberAv3ngers)

Target: IOCONTROL custom ICS malware platform components. Detects structural patterns and strings consistent with IOCONTROL as described by Claroty/Tenable (2024–2026). **Note:** IOCONTROL runs on embedded Linux-based OT devices; scan PLC firmware images and configuration exports. **False Positives:** Low - protocol strings are specific to this malware's ICS-targeting logic. **MITRE ICS ATT&CK;:** T0816 (Device Restart/Shutdown) | T0831 (Manipulation of Control)

```
rule CyberAv3ngers_IOCONTROL_ICS {
    meta:
        description = "Detects IOCONTROL custom ICS malware (CyberAv3ngers/Bauxite)"
        author      = "Transilience AI CTI"
        date        = "2026-04-15"
        reference   = "Claroty/Tenable IOCONTROL Analysis 2024; CISA AA26-097A"
        mitre_ics   = "T0816, T0831"
        tlp         = "CLEAR"
    strings:
        $s1  = "iocontrol"            ascii nocase wide
        $s2  = "/register"            ascii
        $s3  = "/heartbeat"           ascii
        $s4  = "Unitronics"           ascii nocase
        $s5  = "Allen-Bradley"        ascii nocase
        $s6  = "/dev/ttyS"            ascii
        $s7  = "You have been hacked" ascii wide nocase
        $s8  = "down with Israel"     ascii nocase
        $s9  = "mqtt://"              ascii nocase
        $s10 = "mosquitto"            ascii nocase
    condition:
        (
            uint32(0) == 0x464C457F or  // ELF magic (Linux OT binaries)
            uint16(0) == 0x5A4D         // PE magic (Windows components)
        ) and
        filesize < 10MB and
        (
            3 of ($s1, $s2, $s3, $s4, $s5, $s6) or
            any of ($s7, $s8) or
            ($s9 and $s10)
        )
}
```

RULE 04 - Rockwell Engineering Software Connecting to External IPs (T0883 / T0885)

Log Source: Sysmon EID 3 (Network Connection) or MDE DeviceNetworkEvents on engineering workstations running Studio 5000. **Note:** Engineering workstations should NEVER initiate outbound connections to non-RFC1918 IP ranges on OT ports. Any hit is high-confidence. **MITRE ICS ATT&CK;:** T0883 (Internet Accessible Device) | T0885 (Commonly Used Port)

```
# Sigma Rule - Rockwell Engineering Software External C2 Connection
title: Rockwell Studio 5000 or RSLinx Connecting to External IP on OT Ports
id: irgc-cyberav3ngers-ot-c2-004
status: stable
description: |
    Detects Rockwell Automation engineering software (Studio 5000 Logix Designer,
    RSLogix 5000, RSLinx) making outbound connections to non-RFC1918 IP ranges
    on OT ports identified in CISA AA26-097A. In legitimate OT environments,
    engineering software communicates only with PLCs on internal networks.
    Any connection to external IPs on these ports is highly suspicious.
    CISA AA26-097A confirmed: actors used Studio 5000 to create accepted
    connections to internet-exposed PLCs from overseas hosting infrastructure.
author: Transilience AI CTI
```

```
date: 2026-04-15
references:
    - https://www.cisa.gov/news-events/cybersecurity-advisories/aa26-097a
mitre_ics:
    - T0883
    - T0885
logsource:
    category: network_connection
    product: windows
detection:
    selection:
        Image|endswith:
            - '\Studio 5000\Logix Designer\Studio5000.exe'
            - '\Rockwell Software\RSLogix 5000\RSLogix5000.exe'
            - '\Rockwell Software\RSLinx\RSLinx.exe'
        DestinationPort|in:
            - 44818
            - 2222
            - 102
            - 22
            - 502
    filter_rfc1918:
        DestinationIp|startswith:
            - '10.'
            - '172.16.'
            - '172.17.'
            - '172.18.'
            - '172.19.'
            - '172.20.'
            - '172.21.'
            - '172.22.'
            - '172.23.'
            - '172.24.'
            - '172.25.'
            - '172.26.'
            - '172.27.'
            - '172.28.'
            - '172.29.'
            - '172.30.'
            - '172.31.'
            - '192.168.'
    condition: selection and not filter_rfc1918
falsepositives:
    - Extremely low in production OT environments - engineering software should
      only communicate with PLCs on internal network segments
level: critical
```

RULE 05 - Rockwell PLC Project File Modified by Non-Engineering Process (T1565)

Log Source: Sysmon EID 11 (File Create) or MDE DeviceFileEvents. **Note:** .ACD and .L5X files are Rockwell PLC project files. Modification by any process outside the approved Studio 5000 path is a critical indicator. **MITRE ICS ATT&CK;:** T1565 (Stored Data Manipulation) -- confirmed impact technique per AA26-097A.

```
# Sigma Rule - Rockwell PLC Project File Modification by Non-Studio5000 Process
title: PLC Project File (.ACD/.L5X) Modified by Non-Engineering Process
id: irgc-cyberav3ngers-plcmod-005
status: stable
description: |
    Detects writes to Rockwell Automation .ACD or .L5X project files by
```

```
    processes other than approved Studio 5000 Logix Designer or RSLogix.
    Per CISA AA26-097A, CyberAv3ngers extracted, modified, and re-deployed
    PLC project files containing ladder logic and control sequences.
    .ACD files contain complete PLC programs including I/O configuration
    and safety logic. Any modification by an unknown process is critical.
author: Transilience AI CTI
date: 2026-04-15
references:
    - https://www.cisa.gov/news-events/cybersecurity-advisories/aa26-097a
mitre_ics:
    - T1565
logsource:
    category: file_event
    product: windows
detection:
    selection:
        TargetFilename|endswith:
            - '.acd'
            - '.ACD'
            - '.l5x'
            - '.L5X'
    filter_legit:
        Image|contains:
            - '\Studio 5000\'
            - '\RSLogix\'
            - '\Rockwell Software\'
    condition: selection and not filter_legit
falsepositives:
    - Legitimate backup or version control tools archiving PLC project files
    - Add backup software paths to filter_legit as needed
level: critical
```

RULE 06 - Dropbear SSH Process on OT Engineering Workstation (T1219)

Log Source: Sysmon EID 1 (Process Creation) or MDE DeviceProcessEvents. **Note:** Dropbear is a lightweight SSH server for embedded Linux. It has no legitimate use on Windows OT engineering workstations. False-positive rate is near-zero in production OT environments. **MITRE ATT&CK;:** T1219 (Remote Access Tools) -- per AA26-097A, Dropbear deployed on victim endpoints for persistent remote access.

```
# Sigma Rule - Dropbear SSH on OT Engineering Workstation
title: Dropbear SSH Process Detected on OT Engineering Workstation
id: irgc-cyberav3ngers-dropbear-006
status: stable
description: |
    Detects execution of Dropbear SSH, a lightweight SSH server/client for
    embedded systems, on Windows OT engineering workstations. Per CISA
    AA26-097A, CyberAv3ngers deployed Dropbear SSH on victim endpoints
    to establish persistent remote access via port 22. Dropbear has no
    legitimate operational use on Windows engineering workstations.
    False-positive rate is near-zero in OT environments.
author: Transilience AI CTI
date: 2026-04-15
references:
    - https://www.cisa.gov/news-events/cybersecurity-advisories/aa26-097a
mitre_attack:
    - T1219
logsource:
    category: process_creation
    product: windows
detection:
```

```
    selection:
        Image|endswith:
            - '\dropbear.exe'
            - '\dbclient.exe'
        CommandLine|contains: 'dropbear'
    selection_name:
        OriginalFileName: 'dropbear'
    condition: 1 of selection*
falsepositives:
    - None expected on OT engineering workstations
    - If Dropbear is found: treat as confirmed compromise until proven otherwise
level: critical
```

12
Proactive Threat Hunting Pivots

Hunting pivots are behavioral, IOC-rotation-resistant, and distinct from the reactive detection rules in Section 11. These pivots will remain valid even if the specific IOCs in Section 10 are rotated, because they target persistent behaviors rather than specific infrastructure.

Pivot	Actor	What to Hunt	Query / Method	Why Rotation-Resistant
01 - Exchange DLL Sideloading	APT34	DLL files resembling Exchange components loaded from non-standard paths by w3wp.exe	`DeviceImageLoadEvents | where FileName contains 'Exchange' and FolderPath !startswith 'C:\\Program Files\\Microsoft\\'`	Tactic persists across all STEALHOOK variants regardless of DLL filename
02 - Cloud Storage via Office Process	MuddyWater	WINWORD, EXCEL, Acrobat making HTTPS connections to TeraBox, OneHub, Sync.com, Egnyte, Backblaze	`DeviceNetworkEvents | where InitiatingProcessFileName in ('WINWORD.EXE','EXCEL.EXE') and RemoteUrl has_any ('terabox','onehub','sync.com')`	Cloud platform preference persists even as specific staging domains rotate
03 - Password Filter DLL on DCs	APT34	New DLL added to HKLM\SYSTEM\CurrentControlSet\Control\Lsa\Notification Packages on a domain controller	`DeviceRegistryEvents | where RegistryKey contains 'Notification Packages' and ActionType == 'RegistryValueSet' --cross-ref with DC asset list`	Structurally deterministic: a DLL MUST be registered in this exact key to harvest credentials via this technique
04 - Anomalous Entra ID Sign-in	APT33	Accounts with >10 sign-in attempts from >3 distinct ASNs within any 24-hour window	`SigninLogs | summarize ASNs=dcount(AutonomousSystemNumber), Attempts=count() by UserPrincipalName, bin(TimeGenerated,1d) | where ASNs > 3 and Attempts > 10`	Pattern persists regardless of which VPN, TOR exit, or proxy ASN APT33 rotates through

05 - Exposed PLCs with Default Creds	Cyber Av3ngers	OT devices responding on TCP/20256 (Unitronics) and TCP/44818 (Rockwell EtherNet/IP) that accept default or no credentials	`Proactive scan: nmap -p 20256,44818 . Attempt auth with known Unitronics defaults. Configuration audit, not a SIEM rule.`	Targets the permanent attack surface; no IOC rotation can remove an unpatched internet-exposed PLC
06 - Azure Per-Victim C2 Tenant	UNC1549	Connections to *.azurewebsites.net or *.blob.core.windows.net hostnames not in approved cloud inventory and first resolved within the past 30 days	`DNS logs: query for *.azurewebsites.net / *.blob.core.windows.net first seen <30 days ago from defence/aerospace subnets. Cross-ref against approved Azure tenant list.`	Azure per-victim tenant registration has been UNC1549s persistent C2 architectural choice since June 2022

13 Remediation & Compensating Controls

Priority actions are specific, sequenced by time window, and assigned to responsible teams. Compensating controls are validated against specific kill chains. Patch efficacy is stated where verified.

IMMEDIATE - Within 24 Hours

OT / ICS Infrastructure	Audit ALL internet-exposed PLCs, HMIs, and SCADA interfaces immediately. Apply CISA AA26-097A vendor mitigations for Rockwell Allen-Bradley (network isolation, authentication enforcement). CVE-2021-22681: NO VENDOR PATCH - network isolation and strong authentication are the only mitigations. For Unitronics Vision Series: change default credentials on TCP/20256 and place behind VPN. Compensating control: block TCP/20256 and TCP/44818 inbound at perimeter firewall. Kill chain mitigated: Kill Chain 2 (CyberAv3ngers) Steps 1-2.
Identity & Access Management	Enable Entra ID Identity Protection with Conditional Access policies blocking sign-ins from TOR exit node IP ranges. Activate sign-in risk policies targeting go-http-client User-Agent from TOR. Subscribe to https://check.torproject.org/torbulkexitlist for dynamic TOR exit node blocking (update weekly). This directly mitigates the confirmed APT33 active password spray campaign. Kill chain mitigated: APT33 TOR spray, Section 11 Rule 01.
SIEM / Detection Deployment	Deploy Sigma Rule 01 (APT33 TOR spray, Entra ID logs) and Rule 02 (MuddyWater LOLBin chain) from Section 11 immediately. Run Section 12 Pivot 03: audit HKLM\SYSTEM\CurrentControlSet\Control\Lsa\Notification Packages on ALL domain controllers for any unrecognised DLL. Any unknown DLL is a psgfilter.dll credential harvester until proven otherwise. Kill chain mitigated: Kill Chain 1 (APT34) Step 5.
Threat Intelligence Ingestion	Ingest all IOCs from Section 10 into TIP and SIEM with priorities: CRITICAL - IOCONTROL, STEALHOOK, psgfilter.dll (active confirmed exploitation); HIGH - BugSleep/MuddyRot, DarkBeatC2, MINIBIKE/MINIBUS, VAXOne (active campaigns); MEDIUM - domain IOCs (shelf life 6-18 months). Set automated blocking for CRITICAL indicators across EDR, email gateway, and proxy.

SHORT-TERM - Within 1 Week

Vulnerability Management	Verify CVE-2024-30088 patch (KB5039212 for Windows 10; KB5039213 for Windows 11) is applied across all endpoints. Patch efficacy confirmed by Trend Micro (October 2024): fully eliminates APT34 primary privilege escalation vector. Prioritise domain controllers and Exchange servers. Compensating control for unpatched systems: block certutil.exe and regsvr32.exe from spawning PowerShell (Sigma Rule 02). Kill chain mitigated: Kill Chain 1 (APT34) Step 3.
Exchange Server Hardening	Audit all Exchange servers for web shells (IIS logs for unusual POST requests), ngrok.exe, and unexpected DLLs loaded by w3wp.exe (Section 12 Pivot 01). Restrict Exchange outbound SMTP to approved domains only - this severs the STEALHOOK exfiltration channel (Kill Chain 1 Step 7). Enable Advanced Audit Policy logging on Exchange. Deploy KQL Rule 03 (Section 11) for RunPE-In-Memory detection. If CVE-2024-30088 cannot be patched immediately, enable Windows Credential Guard on domain controllers as compensating control.
OT Network Segmentation	Implement network segmentation isolating OT/ICS from IT. All OT devices must be on isolated VLANs with deny-all ingress/egress as default. Block MQTT (TCP/1883, TCP/8883) outbound from OT segments - IOCONTROL uses MQTT for C2. Apply CISA AA26-097A recommended network controls: PLC communication permitted only from known engineering workstations on isolated segments.
Email Security	Enable DMARC enforcement (p=reject) for all owned domains to block APT35 spoofed-domain spearphishing. Block or alert on outbound HTTPS connections to TeraBox, OneHub, Sync.com, and Egnyte from managed endpoints (MuddyWater cloud staging, Section 12 Pivot 02). Enable URL detonation for all external email links. Kill chain mitigated: Kill Chain 3 Step 1.
Supply Chain / Third-Party	Audit all third-party IT providers and MSPs with administrative access. Require FIDO2 hardware MFA and privileged access workstations (PAW) for all remote vendor connections. Review access logs for 90 days for anomalous vendor activity. Reference: MuddyWater Rashim Software supply-chain compromise (Kill Chain 3 Step 6). Rashim downstream customers at elevated cascade risk.

MEDIUM-TERM - Within 30 Days

Phishing-Resistant MFA Rollout	Enrol all high-risk accounts (senior officials, IT admins, policy staff, executives) in FIDO2 hardware security keys. APT35 MFA interception kits specifically defeat TOTP (Google Authenticator, Microsoft Authenticator) and SMS-based MFA - confirmed by DOJ Sep 2024 indictment. Hardware keys provide chain-specific mitigation that intercept kits cannot defeat. TOTP is insufficient for accounts with access to sensitive material. Kill chain mitigated: APT35 T1111 MFA interception.

Credential Exposure Assessment	Run a full Active Directory credential exposure assessment using BloodHound or equivalent. Identify all paths to domain compromise. Audit all DC LSA Notification Packages registry values. Rotate all privileged account passwords. Check all service accounts for Kerberoastable SPNs. If psgfilter.dll was present on any DC: treat ALL domain credentials processed by that DC as compromised and force-rotate them. Kill chain mitigated: Kill Chain 1 Steps 5-6.
Security Awareness Training	Deploy targeted spearphishing simulation using APT35-documented lure types from Section 06: conference invitations, speaking engagement requests, embassy event invitations, conflict-themed job offers. Train staff: rapport-building lures may precede the malicious link by weeks. Verify sender identity via secondary channel before clicking any external link. Kill chain mitigated: APT35 social engineering KC.
OT Backup & Recovery	Validate backup integrity for all PLC project files and OT configurations. Test restoration procedures for Unitronics and Rockwell PLCs. Establish offline air-gapped backups of all PLC project files updated at least monthly. Pre-authorise an OT/ICS-capable IR retainer. Document manual operating procedures for all critical processes that could be disrupted by IOCONTROL. Kill chain mitigated: Kill Chain 2 (CyberAv3ngers) Step 4 recovery.
IR Tabletop Exercise	Conduct tabletop exercise using Section 14 playbook against two scenarios: (1) CyberAv3ngers IOCONTROL on PLCs; (2) APT34 psgfilter.dll on domain controllers. Pre-authorise notification to CISA, national CERT, and legal/DPO. Establish war-room communication channels and escalation matrix. Verify backup media is physically accessible and not connected to any potentially compromised segment.

14
Incident Response Playbook

This playbook applies to active exploitation scenarios, specifically: IOCONTROL on OT devices, STEALHOOK on Exchange servers, and BugSleep persistence. Sequence follows: ISOLATE → MEMORY → IMAGE → ARTIFACTS → CREDENTIALS → LEGAL → REPORT.

1. ISOLATE	Immediately segment affected systems from network. For OT/PLC devices: remove from internet; revert to manual/local control. For compromised Exchange servers: isolate from external mail relay. Preserve volatile state - do NOT power off before memory acquisition.
2. MEMORY ACQUISITION	Acquire full memory dump of compromised systems using validated tools (WinPmem, Magnet RAM Capture). For OT devices: follow vendor-specific memory acquisition procedures. Hash all memory images immediately.
3. DISK IMAGE	Create forensic disk images of all affected systems. Use write blockers. Verify SHA-256 hashes post-acquisition. Do not run forensic analysis on originals.
4. ARTIFACT COLLECTION	Collect: Windows Event Logs (Security, System, Application, PowerShell), Sysmon logs, Exchange transport logs, IIS logs, PLC project file backups, network PCAP if available. Export Entra ID sign-in logs for the 90-day retention window immediately.
5. CREDENTIAL RESET	Reset ALL domain controller service accounts and domain admin credentials immediately - psgfilter.dll may have captured them. Reset Entra ID/Azure AD service principal secrets. Rotate API keys for cloud services. Enable forced re-authentication for all sessions.
6. LEGAL / DPO NOTIFICATION	Assess notification obligations per Section 15. Notify DPO. Preserve attorney-client privilege for investigation communications. Do not delete potentially relevant data.
7. REPORT / NOTIFY	Report to CISA (cisa.gov/report) if critical infrastructure. Report to national CERT. Notify IRGC Cyber-Electronic Command activity to FBI IC3 (ic3.gov) for correlation. Issue internal notification to board and senior leadership.

15
Legal & Regulatory Notification Assessment

IRGC cyber operations resulting in unauthorized access to personal data or critical infrastructure may trigger mandatory notification obligations. This section maps confirmed attack scenarios to applicable regulatory frameworks.

Regulation	Trigger Scenario	Notification Deadline	Recipient	ME Applicability
GDPR (EU)	APT35/MuddyWater credential theft affecting EU citizen data	72 hours to DPA after discovery	Supervisory Authority (lead DPA)	Orgs with EU operations or EU citizen data
DPDP Act (India)	Breach affecting Indian citizens' personal data	6 hours to CERT-In (CRITICAL INFRA STRUCTURE ONLY - this deadline applies to critical sector entities per CERT-In directions; non-critical-infra organisations follow standard data fiduciary notification timelines as prescribed by the Data Protection Board once operationalised)	CERT-In; Data Protection Board of India	Indian operations / data processors
PCI-DSS v4.0.1	Payment card environment compromise (energy/retail sectors)	Immediately to acquirer/card brands	Payment brands (Visa, Mastercard)	GCC merchants and processors
DORA (EU)	ICT-related incident affecting EU financial entity	4 hours (initial); 72 hours (intermediate); 1 month (final)	Lead financial supervisor + ENISA	Financial entities with EU operations

UAE PDPL	Breach of UAE resident personal data	72 hours to UAE Data Office	UAE Data Office	UAE operations
Saudi PDPL	Breach of Saudi resident personal data	72 hours to SDAIA	SDAIA	Saudi Arabia operations
CIRCIA (USA)	Substantial cyber incident or ransomware payment affecting U.S. critical infrastructure -- directly applicable to Water/Wastewater, Energy, Government sectors targeted by AA26-097A	72 hours (substantial incident); 24 hours (ransom payment). NOTE: Final rule still pending as of April 2026; CISA town halls March 2026, final rule expected May 2026. Mandatory reporting NOT yet in effect -- voluntary reporting strongly encouraged via cisa.gov/report	CISA	U.S. critical infrastructure operators (16 sectors)
NERC CIP (USA)	Cyber incident affecting Bulk Electric System (BES) cyber assets -- internet-exposed PLCs controlling generation or transmission assets trigger CIP-005 (Electronic Security Perimeter) and CIP-007 (System Security Management) violations	Report to E-ISAC within 1 hour (Reportable Cyber Security Incident per CIP-008-6); full report within 7 calendar days	NERC E-ISAC + FERC	Bulk electric system operators, generation and transmission owners (U.S. and Canada)

16 Business Impact Assessment

Business impact is quantified using confirmed victim counts, data categories at risk, and estimated dwell time ranges from published reports.

QUANTIFIED IMPACT BY CLUSTER

Cluster	Installed Base at Risk	Data Categories	Estimated Dwell Time	Financial Impact
CyberAv3ngers	75+ PLCs confirmed compromised (2023); 400+ OT devices via IOCONTROL (Tenable Apr 2026); 60+ groups now using similar techniques (Tenable Apr 2026; single commercial source)	OT/ICS operational data; process setpoints; HMI display data	Days to weeks (disruptive, not dwell-focused)	Confirmed operational disruption and financial losses (CISA AA26-097A)

APT34/OilRig	Multiple government and energy orgs in UAE, Gulf; supply chain access via IT providers	Government communications, credentials, financial data, strategic documents	Months to years (long-term persistence objective). Analogous reference: Fox Kitten campaign (2017-2020, APT33+APT34 VPN exploitation, Section 17 Phase 2) documented dwell times of 12-36 months in target organisations before detection. ESET OilRig 'Out to Sea' campaign (2018-2022) sustained access across the same targets for over three years. Clients in Gulf energy or government should treat APT34 dwell times as measured in years, not weeks.	Not directly quantified in public reporting for current campaigns; financial impact is intelligence-loss and pre-positioning value, not operational disruption. See dwell time analogues from Phase 2 historical comparison (Section 17).
APT35	Thousands of individuals targeted (DOJ indictment); multiple US political campaign members; ME affairs personnel	Email content, credentials, travel/location data, policy positions, PII	Ongoing account access after initial compromise; some accounts monitored for years	Election influence operations; reputational harm to targets; ongoing state surveillance risk

MuddyWater	Local authorities, healthcare, telecom, IT in Israel and Egypt; Rashim IT supplier (cascade unknown)	Network credentials, browser data, enterprise documents, customer data via compromised IT providers	Weeks to months	Not quantified; supply-chain access multiplies downstream exposure

Downstream / Dependency Impact: MuddyWater's compromise of the Rashim IT provider demonstrates IRGC willingness to exploit managed service provider access for cascade targeting. Organizations using third-party IT management, cloud migration services, or remote monitoring from Middle Eastern MSPs are at elevated downstream risk. The specific number of downstream Rashim customers affected is NOT publicly disclosed - this is flagged as an assessed but unquantified downstream exposure.

17 Historical Comparison

Iran's state-directed cyber programme has undergone three distinct evolutionary phases over fifteen years. Understanding this trajectory is essential for anticipating how capabilities will continue to develop and for identifying which historical patterns remain operationally relevant today.

PHASE 1: DISRUPTION AND DEFACEMENT (2010-2015)

Origins (2009-2010)	Iran's offensive cyber programme emerged from the 2009 Green Movement. The Iranian Cyber Army's redirection of Twitter traffic (Dec 2009) and defacement of Baidu (Jan 2010) established the foundational deniable-proxy model: operations presented as independent hacktivist activity while coordinated under IRGC oversight. This template is the direct ancestor of CyberAv3ngers and the June 2025 hacktivist ecosystem.
Operation Ababil Dec 2011 - May 2013	DDoS campaign against 46 U.S. financial institutions including Bank of America, JPMorgan Chase, Wells Fargo, and the NYSE. Seven IRGC-linked individuals indicted in absentia (DOJ, March 2016). One co-conspirator also accessed SCADA systems of the Bowman Avenue Dam, Rye, New York (Aug-Sep 2013) - the first confirmed IRGC ICS intrusion on U.S. soil.
Shamoon / Saudi Aramco Aug 15, 2012	IRGC-linked Cutting Sword of Justice deployed the Shamoon/Disttrack wiper, deleting and overwriting data on approximately 35,000 computers. Timed to Ramadan when staff were absent. The NSA: 'Iran's destructive cyberattack against Saudi Aramco in August 2012 was the first such attack NSA has observed from this adversary.' Shamoon was itself adaptive - Iran had suffered the Wiper malware attack against its own oil ministry in April 2012, studied it, and weaponised it.
Sands Casino Feb 2014	Destructive attack against Las Vegas Sands Corporation directly attributed to the Iranian government by the U.S. Director of National Intelligence. Deployed as retaliation for the owner's public anti-Iran political statements - establishing the doctrinal pattern of destructive capability as a geopolitical retaliation signal.
Phase 1 Doctrinal Legacy	Three enduring doctrines established: (1) deniable proxy fronts masking state direction; (2) destructive wiper operations as geopolitical retaliation signals; (3) ICS/OT networks as a preferred target domain, demonstrated years before CyberAv3ngers by the Bowman Dam SCADA intrusion.

PHASE 2: APT MATURATION AND ESPIONAGE SCALING (2015-2022)

APT Professionalisation 2014-2017	APT34/OilRig active since at least 2014, targeting Gulf government, financial, energy, and chemical sectors. Operation Saffron Rose (2013-2014, APT35 predecessor Ajax Security Team) established the dual-track model: strategic espionage plus targeted harassment of Iranian dissidents and U.S. defence personnel - the same model confirmed by the DOJ Sep 2024 indictment.
Shamoon Returns 2016-2017	A second wave of Shamoon attacks struck Saudi Arabia's General Authority of Civil Aviation, Ministry of Labor, and Saudi Central Bank. Files overwritten with the image of a drowned Syrian refugee, demonstrating evolving psychological operations alongside technical destruction. Linked to StoneDrill spyware in early 2017 (Symantec).
Thamar Reservoir 2014-2015	ClearSky identified a widespread phishing operation targeting 500+ academic researchers in Israel, Saudi Arabia, and other ME countries - active since at least 2011. Established APT35's academic and policy-community targeting as a sustained strategic priority.
Fox Kitten Campaign 2017-2020	APT33 and APT34 collaborated on systematic VPN appliance exploitation: Pulse Secure CVE-2019-11510, Fortinet CVE-2018-13379, Palo Alto GlobalProtect. Enabled simultaneous persistent footholds across IT, telecom, oil and gas, aviation, and government organisations globally. Infrastructure explicitly assessed as convertible for large-scale wiper deployment (ClearSky, Feb 2020).
MuddyWater Emergence 2017 onwards	Debuted with PowerShell-heavy tradecraft and LOLBin abuse (POWERSTATS backdoor). The POWERSTATS > MuddyC3 > PhonyC2 > MuddyC2Go > DarkBeatC2 framework evolution over seven years reflects sustained institutional R&D; investment. NCSC/GCHQ formally attributed to MOIS in January 2022.
Phase 2 Doctrinal Legacy	Three advances: (1) scalable VPN-based initial access as a repeatable inter-group playbook; (2) confirmed APT33 + APT34 operational collaboration on shared infrastructure; (3) pre-positioned destructive capability held in reserve for strategic activation - a posture now directly manifest in UNC1549's ICS terrain-mapping in 2025-2026.

PHASE 3: HYBRID OPERATIONS AND ICS ESCALATION (2022-PRESENT)

Shift 1: ICS/OT Physical Effects 2023-present	CyberAv3ngers' 2023 PLC defacement was reconnaissance and capability-testing, not an end state. IOCONTROL - a custom ICS malware platform supporting six PLC/HMI vendor families - confirmed state-directed OT weapons development. CISA AA26-097A (Apr 2026) confirmed active CVE-2021-22681 exploitation with no patch available, causing confirmed operational disruption and financial losses. IOCONTROL deployed to 400+ OT devices globally.
Shift 2: Cloud as C2 2022-present	All major clusters migrated from dedicated servers to legitimate cloud platforms: APT34 uses Exchange Online and OneDrive; UNC1549 registers per-victim Azure tenants; MuddyWater cycles through TeraBox, OneHub, Sync.com, and Egnyte; APT35 uses Dropbox, Google Drive, Backblaze, and IPFS. Exploits implicit organisational trust in cloud traffic and defeats network-layer blocking.
Shift 3: Hacktivist Integration Jun 2025	The twelve-day Iran-Israel conflict demonstrated state APT / proxy integration at unprecedented scale. SecurityScorecard analysis of 250,000+ Telegram messages from 178+ groups confirmed coordinated target selection, shared exploit scripts, and timing correlated with kinetic military operations. SEPAHCYBERY channel ran parallel psychological operations. CyberAv3ngers ICS techniques proliferated to an estimated 60+ affiliated groups (Tenable Apr 2026; single commercial source).
Shift 4: Attribution Architecture 2025-present	The KittenBusters data leak (Sep-Oct 2025) exposed the complete IRGC Unit 1500 command structure: Bitcoin-funded VPS procurement confirmed, Shuhada base in Tehran as operational HQ, and direct administrative overlap between CyberAv3ngers and the Moses Staff persona. Iran's deniability now requires leaked insider documents to fully characterise.
Current Assessment Apr 2026	Following Feb 28 U.S.-Israel strikes on IRGC facilities, all clusters operate at elevated retaliatory posture (CloudSEK, Mar 2026; Nozomi, Mar 2026). 61% of vulnerabilities in ME-region organisations are HIGH or CRITICAL vs. 48% globally. Iran is no longer pre-positioning for future ICS operations - it is executing them at scale, against unpatched targets, with techniques distributed to an estimated 60+ affiliated groups (Tenable Apr 2026; single commercial source) as a force-multiplication strategy.

COMPARATIVE CAPABILITY TRAJECTORY

Dimension	2015	2020	2026	Direction
Initial Access	Spearphishing, web shells	VPN exploitation, supply chain	Password spraying at cloud scale; ICS scanning; MSP compromise	Broadening

C2 Infrastructure	Dedicated attacker servers	Legitimate cloud storage (Dropbox, Drive)	Per-victim Azure tenants; Telegram API; DarkBeatC2; ICS-native protocols	**Hardening**
Destructive Capability	Shamoon wiper (Saudi Aramco)	PowGoop wiper (Israel); OT defacement	IOCONTROL ICS malware; PLC manipulation with physical effects; wiper + ransomware hybrid	**Escalating**
Proxy Ecosystem	Small number of aligned hacktivist groups	Moses Staff; CyberAv3ngers; Homeland Justice	60+ groups using proliferated ICS techniques (Tenable Apr 2026; single-source); 178+ hacktivist channels coordinated in June 2025	**Scaling rapidly**
Target Sectors	Government, energy, dissidents	Defence, aviation, telecom, water utilities	All prior sectors plus ICS/OT across water, energy, manufacturing; journalists; election infrastructure	**Widening**
Dwell Time	Days to weeks	Weeks to months	Months to years; pre-positioning in OT environments without triggering effects	**Lengthening**
Attribution Deniability	Weak - operations linked to state through infrastructure	Moderate - hacktivist personas used	Strong - layered proxy ecosystem; Bitcoin-funded VPS; front companies managing operations	**Improving**

Methodology note: *Capability assessments are derived from published government advisories (CISA, FBI, NSA), commercial threat intelligence (Mandiant, Picus Security, Nozomi Networks, Tenable), and independent analysis (CSIS, SecurityScorecard). Year columns represent the state of capability at the end of the indicated year based on publicly documented operations. The 2026 column reflects the current assessment as of April 15, 2026.*

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Threat Actor Motivation Assessment

Motivation is assessed independently from attribution. See Section 04 for detailed per-cluster motivation analysis. This section provides the consolidated motivation taxonomy.

Motivation Category	Clusters	Evidence
Strategic Espionage	APT33, APT34, APT35, UNC1549	Long-dwell intrusions; targeting government, policy, and defence organizations; data categories consistent with intelligence collection (credentials, strategic docs, travel data)
Targeted Harassment & Transnational Repression	APT35, Banished Kitten	DOJ indictment: targeting of Soleimani-assassination-linked individuals; CPJ report: journalist Telegram compromise for intimidation
Operational Disruption (Critical Infrastructure)	CyberAv3ngers, MuddyWater (secondary)	CISA AA26-097A: confirmed operational disruption and financial losses at water/energy/government orgs; PowGoop wiper
Influence Operations	APT35, hacktivist proxies	DOJ: hack-and-leak targeting 2024 U.S. elections; SecurityScorecard: psychological operations via SEPAHCYBERY Telegram channel
Pre-positioning for Future Activation	UNC1549/Pyroxene	HSToday March 2026: ICS kill-chain Stage 2 operations mapping OT pathways in supplier networks - not yet causing disruption

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Sources & References

20+ independent sources across three source types: government advisories, commercial threat intelligence, and academic/think-tank/NGO analysis. Most recent sources prioritised. At least one independent analysis per major attribution claim.

Government / Regulatory

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Commercial Threat Intelligence

[7] Trend Micro - Earth Simnavaz (APT34) Levies Advanced Cyberattacks Against Middle East, October 2024 [link]

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END OF REPORT

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CLASSIFICATION	CONTACT	VERSION
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